

Recommendation Systems

4/12/2026

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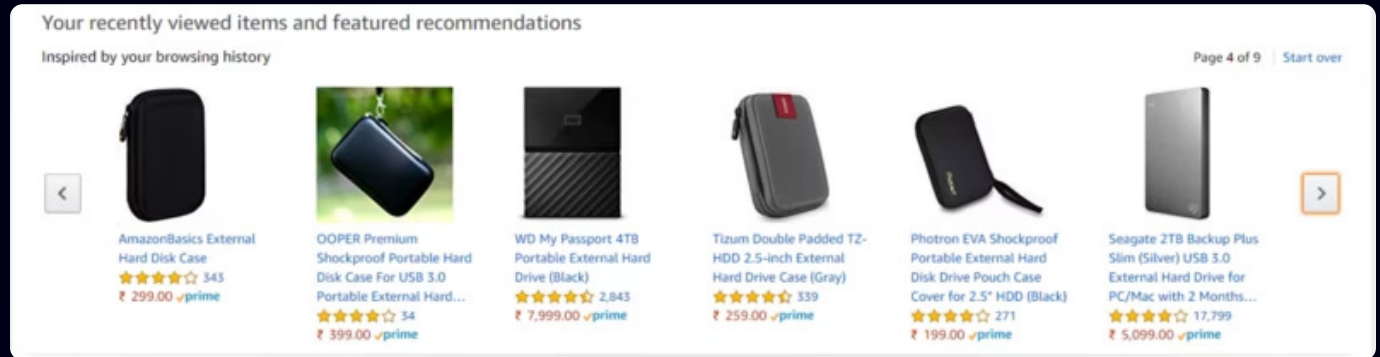
- ☒ **This item:** Samsung Galaxy A8+ (Black, 6GB RAM, 64GB Storage) with Offers ₹23,990.00
- ☒ Spigen Liquid Air Case for Samsung Galaxy A8 Plus (2018) - Matte Black 591CS22757 ₹949.00

Attendance - "Recommender"



What is a Recommender System?

Algorithms that predict user preferences and suggest relevant items by analyzing past behavior and patterns.



Netflix

Movie and show suggestions based on viewing history



Amazon

Product recommendations aligned with your interests



Spotify

Custom music playlists curated to your taste

Explicit feedback vs implicit feedback

1

Explicit feedback

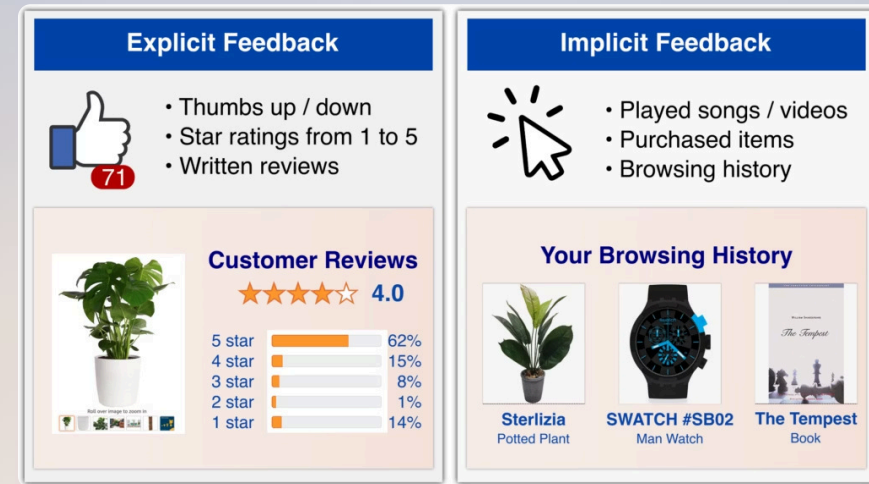
Rating explicitly given by the user to express their satisfaction with an item

2

Implicit feedback

Assume that user-item interactions are an indication of preferences

Most modern recommender systems tend to rely on implicit feedback



Process

1

Data collection

Collecting explicit and implicit feedback

2

Data storage

Storing data in structured systems

3

Data analysis

Using machine learning models to identify data patterns and correlations across large datasets

4

Data filtering

Filtering to the most relevant items for each particular user

5

Feedback loop

Evaluating the recommendation system's outputs to improve the model

Types of Recommender Systems



Collaborative Filtering

Recommends items based on the behavior and preferences of similar users. "Users like you also liked..."



Content-Based Filtering

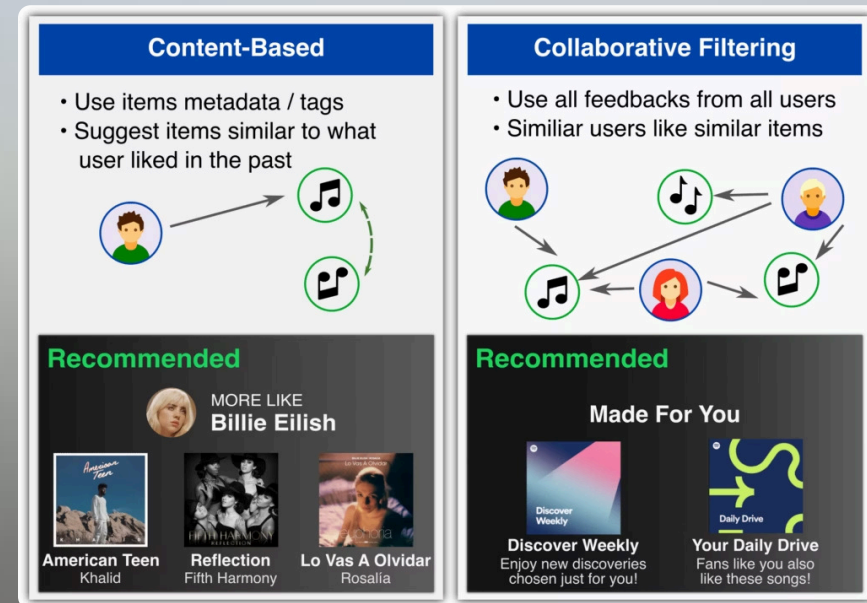
Recommends items based on their features and the user's past preferences. "Because you liked this..."



Hybrid recommenders

Combine benefits of both collaborative and content-based filtering

These algorithms are implemented in the data filtering phase of the recommender system



Content-Based Filtering

Key Principle

Focuses on the characteristics of items and how they match a user's profile

How It Works

Recommends items similar to those a user has liked or interacted with before

Individual Focus

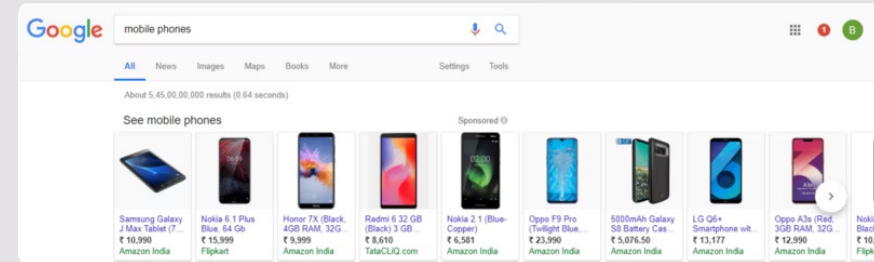
Does NOT rely on data from other users

Feature-Driven

Analyzes item characteristics and attributes

Personal Profile

Builds unique preference profile for each user



More details

1

Embeddings

Items (and the user) are represented as vector embeddings

2

Similarity metrics

Cosine similarity

Euclidean distance

Dot product

3

Evaluation metrics

Precision@K and Recall@K

NDCG (normalized discounted cumulative gain)

MAP (mean average precision)

CTR and conversion rate

RMSE and MAE

Similarity

	Children's Lit	Adventure	Bidungsroman
Little Women	1	0	1
Northanger Abbey	0	0	1
Peter Pan	1	1	0
Treasure Island	1	1	0

One-hot encoding

Items that fit in a particular category have a value of 1 while items that do not fit have a value of 0. This is used to create a vector for each item

$$\text{Cosine}(x,y) = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

Cosine similarity

This calculates a value for how close two vectors are to each other based on the angle between them. A value of 1 is the most alike while -1 is the least alike

Advantages of Content-Based Filtering



Transparency

Recommendations are explainable based on item features and user preferences



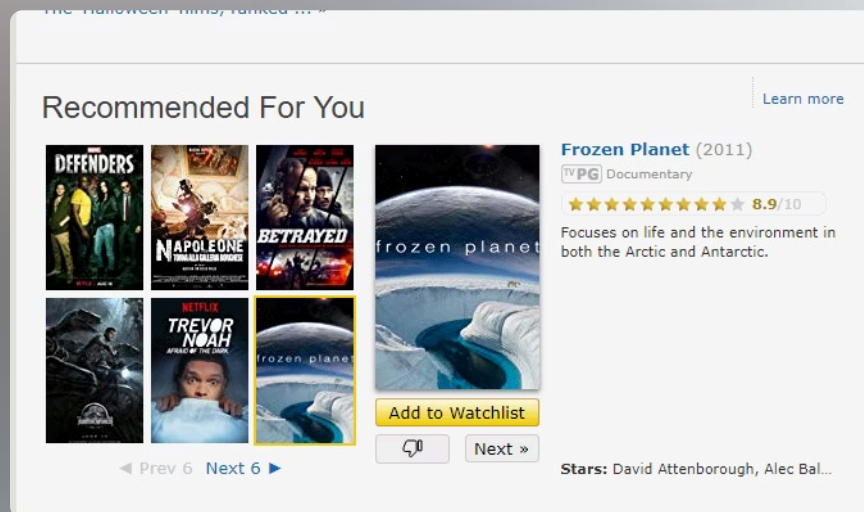
Personalization

Tailored to individual user interests without influence from other users



Reduced Cold Start for Items

Can recommend new items if their features are well-defined and documented



Challenges of Content-Based Filtering

Over-Specialization

May lead to a lack of diversity in recommendations, creating filter bubbles and limiting exposure to new categories.

Feature Engineering

Requires significant effort and domain expertise to define and extract effective item features that truly capture user preferences.

Content-Based Filtering in Action



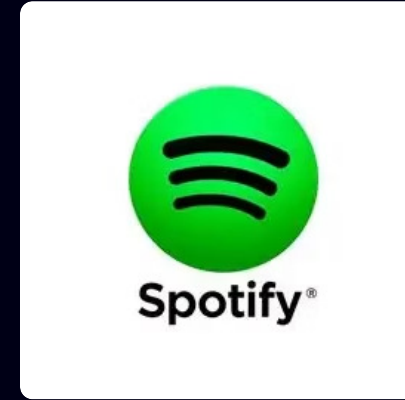
E-commerce

Amazon recommending similar products based on features like category, brand, and price range



Streaming Services

Netflix suggesting movies based on genre, actors, plot keywords



Music Platforms

Spotify recommending songs by genre or artist

- ❏ **Real-World Practice:** These systems are often combined with collaborative filtering in hybrid systems for better results and improved user experience.

Google Colab



Elections Form



Thank you for coming!

Next meeting: 4/19 - Recommender Systems Part 2